ARTIFICIAL INTELLIGENCE (100404)

UNIT -I
INTRODUCTION TO ARTIFICIAL INTELLIGENCE

CO-1 Explain the basic principles and scope of Artificial Intelligence and its relevance to different engineering disciplines

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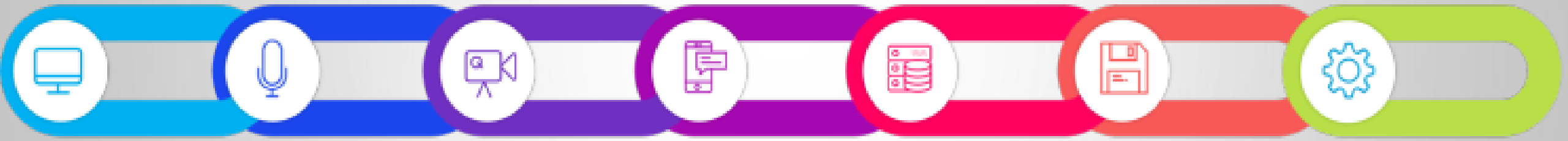
ARTIFICIAL INTELLIGENCE

- Artificial Intelligence (AI) is the **ability of a computer system to perform tasks that normally require human intelligence**. These tasks include:
  - Learning (acquiring knowledge)
  - Reasoning (solving problems)
  - Perception (understanding vision, speech)
  - Natural Language Processing (understanding human language)

SCOPE OF AI

Industry	The Scope of AI	Examples
Technology	Expansive scope of AI in crunching numbers, reading data and providing actionable insights.	Cloud computing, NLPs, ML modules, etc., are used for everyday tech like virtual assistants. (Siri, Alexa)
Healthcare	Diagnosis, screening and treatment procedures become more streamlined and accurate based on past data.	AI-powered robotic surgical equipment improves surgical precision during operations, minimizing the chances of human error and reducing recovery time.
Finance	Increased scope of Artificial Intelligence in reading market data, predicting future money trends and investment plans.	Investment bankers can use AI to predict future risks and provide accurate investment advice to their clients.
Education	Streamlining and automating educator tasks and freeing them up for big-picture thinking. Another scope of AI includes online learning too.	The scope of AI in an online classroom could show up as- real-time speech to text translations, accurate attendance systems and more.
Environment	Sustainable problem-solving through actionable and data-driven insights into the environmental challenges.	Reading soil patterns to help farmers plant the right crops and increasing crop yield.

History of AI



1940 - 1950s
Foundation of AI

In the 1940s, the first artificial neurons were conceptualized. The 1950s introduced us to the Turing Test and the term "Artificial Intelligence."

1960 - 1970s
Early Development

The 60s and 70s brought the birth of ELIZA, simulating human conversation, and Dendral, the first expert system, showcasing theearly potentials of AI.

1980s
AI Winter and Expert Systems

The 80s faced reduced AI funding but saw the inaugural National Conference on AI. The backpropagation concept rejuvenated neural networks.

1990s
Revival and Emergence of ML

The 90s witnessed IBM's Deep Blue defeating chess champion Garry Kasparov. Loom project that is GenAI foundation

2000s
The Genesis of Generative AI

Geoffrey Hinton propelled deep learning into the limelight, steering AI toward relentless growth and innovation.

2010s
Rise of AI

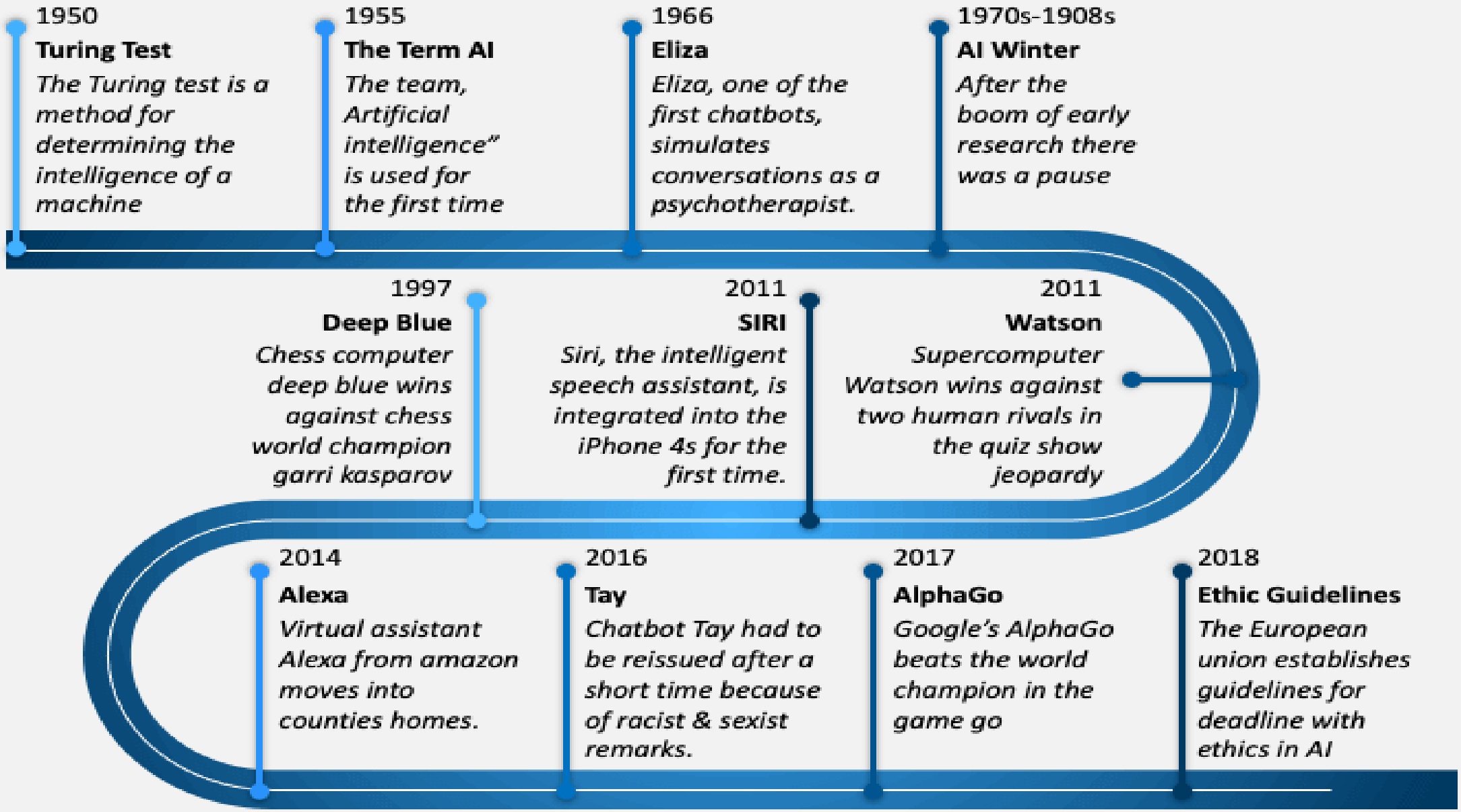
2010s marked major AI milestones, including pioneering work in image recognition, NLP and the birth of GANs in 2014, followed by OpenAI's founding in 2015.

2020s
GenAI Reaches New Horizons

OpenAI's GPT-3 and DALL-E. 2023 welcomed advanced tools like ChatGPT-4 and Google's Bard,and Gemini alongside Microsoft's Bing AI.

HISTORY OF ARTIFICIAL INTELLIGENCE (AI)

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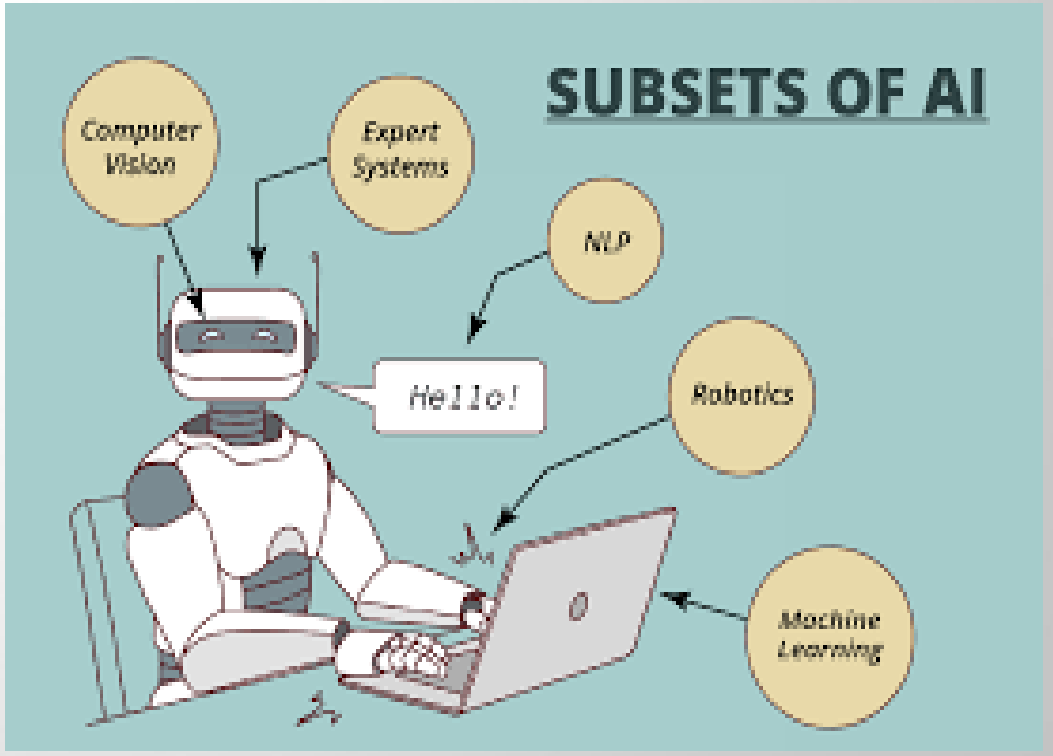


GOALS OF ARTIFICIAL INTELLIGENCE



AI TECHNIQUES

- AI techniques are the methods and algorithms that enable machines to perform tasks that typically require human intelligence.
- These techniques, such as machine learning, deep learning, and natural language processing, allow AI systems to learn, adapt, and solve problems.
- Some of the widely used ones are:
  1. **Machine Learning.**
  2. **Natural Language Processing.**
  3. **Computer Vision.**
  4. **Deep Learning**
  5. **Robotics.**



MACHINE LEARNING (ML)

- Machine Learning is an AI technique **that focuses on the development of computer algorithms that improve automatically through experience and by the use of data**.
- In simpler terms, machine learning enables computers to learn from data and make decisions or predictions without being explicitly programmed to do so..

Types of Machine Learning:

- **Supervised Learning:** Learns from labeled data.
  Example: Email spam detection. The system is trained with examples of spam and non-spam emails.
- **Unsupervised Learning:** Learns patterns from unlabeled data.
  Example: Customer segmentation in marketing (grouping similar customers without prior labels).
- **Reinforcement Learning:** Learns by trial and error using rewards and punishments.
  Example: AI playing games like chess or self-driving cars improving by experience.

NATURAL LANGUAGE PROCESSING (NLP)

- NLP is an AI technique that allows machines to **understand, interpret, and generate human language.**
- **Examples:**
  - **Chatbots:** AI can hold conversations (e.g., customer support bots).
  - **Language Translation:** Google Translate uses NLP to translate languages.
  - **Voice Assistants:** Siri and Google Assistant use NLP to process voice commands.

DEEP LEARNING

- Deep Learning is a **subfield of machine learning** that uses neural networks with multiple layers to mimic the human brain.

Examples:

- **Image Recognition:** AI identifies objects, like recognizing faces on Facebook.
- **Speech Recognition:** Voice assistants like Siri and Alexa use deep learning to understand speech.

COMPUTER VISION

- Computer Vision is the AI technique that allows machines to **see, analyze, and understand images and videos.**
- **Examples:**
  - **Face Recognition:** Used in phone unlocking and security systems.
  - **Self-Driving Cars:** Detect traffic signs, pedestrians, and other vehicles.

ROBOTICS

- Robotics combines AI with mechanical devices to **build robots that can perform tasks independently.**
- **Examples:**
  - **Assembly Line Robots:** Used in car manufacturing.
  - **Service Robots:** Used in hotels, hospitals, and homes to assist people.

HOW AI WORKS

- **Step 1: Data Collection**
  - AI starts with data.
  - It could be text, images, videos, voice, numbers, etc.
  - Example: A self-driving car collects data from sensors and cameras.
- **Step 2: Data Preparation**
  - Raw data is cleaned and organized for the AI model.
  - Remove errors, fill missing values, and convert data into a usable format.
  - This step is called **Data Preprocessing**.
- **Step 3: Choose the Right Algorithm**
  - An AI system uses a specific **algorithm** depending on the task:
  - For prediction: **Machine Learning (ML)**
  - For conversation: **Natural Language Processing (NLP)**
  - For recognizing images: **Computer Vision**
  - For planning/decision making: **Reinforcement Learning**

HOW AI WORKS

- **Step 4:Training the Model**
  - This is the learning phase.
  - The model is given **input data** and **correct output** (called labeled data).
  - It **learns patterns** by adjusting its internal settings (called weights).
  - The process continues until it performs well.
- **Step 5:Testing the Model**
  - The trained model is now tested on **new data** (that it hasn’t seen before).
  - Purpose:To check how well it can make predictions.
  - This is called **model evaluation**.

HOW AI WORKS

- **Step 6: Deployment**
  - Once the model performs well, it is **deployed in the real world.**
- **Step 7: Feedback & Improvement**
  - AI keeps learning and improving from new data and user feedback.
  - This is called **model retraining** or **continuous learning.**

HOW AI WORKS - EXAMPLE

- **Step 1: Data Collection**
- Collect thousands of emails: Some labeled as Spam, some as Not Spam (Ham)
- Data includes:
    - Subject line
    - Email body text
    - Sender address
    - Attachments
- Example data:

Email Text	Label
"Congratulations! You won a prize"	Spam
"Meeting at 10 AM tomorrow"	Not Spam
"Limited offer, click now"	Spam

HOW AI WORKS- EXAMPLE

- **Step 2: Data Preparation**
- **Clean** the emails:
  - Remove stopwords (like "is", "the", "and")
  - Convert all text to lowercase
  - Remove punctuation/symbols
- **Tokenize** (break text into words)
- Convert words to **numerical values** using techniques like:
  - Bag of Words (BoW)
  - TF-IDF (Term Frequency-Inverse Document Frequency)
- Split data into:
  - **Training set** (80%)
  - **Testing set** (20%)

HOW AI WORKS- EXAMPLE

- **Step 3: Choose the Right Algorithm**
- Use a **Machine Learning algorithm** suitable for classification:
  - Example: **Naive Bayes**, **Logistic Regression**, or **Decision Tree**
- Goal: Classify each email as **Spam** or **Not Spam**

HOW AI WORKS- EXAMPLE

- **Step 4: Train the Model**
- Input: Labeled training data
- The model **learns** patterns:
  - Words that appear frequently in spam emails (e.g., "free", "winner", "offer")
  - Words more common in genuine emails (e.g., "project", "meeting", "schedule")

HOW AI WORKS - EXAMPLE

- **Step 5: Test the Model**
- Feed the **testing set** to the model (emails it hasn't seen before)
- Check how accurately it predicts **Spam vs Not Spam**
- Evaluate using metrics:
  - Accuracy
  - Precision
  - Recall
  - Confusion Matrix

HOW AI WORKS -EXAMPLE

- **Step 6: Deploy the Model**
- Integrate the model into the email server or spam filter system
- Each incoming email is passed to the model
- It gives output:
  - **Spam → Move to spam folder**
  - **Not Spam → Inbox**

HOW AI WORKS - EXAMPLE

- **Step 7: Feedback & Improvement**
- Users can mark an email as **“Not Spam”** if wrongly classified
- System logs such feedback
- Periodically, retrain the model using:
  - New emails
  - User corrections
- Model gets **smarter over time**

APPLICATION OF AI

- Google Predictive Search Engine
- JP Morgans Chase’s Contract Intelligence Platform
- Googles AI EYE Doctor
- Social Medial Platforms like facebook
- Twitters AI
- Virtual Assistants
- Tesla's Self Driving Cars
- Netflix

HEALTHCARE

- Disease Diagnosis: AI helps in detecting diseases faster and more accurately.
- Medical Imaging: AI analyzes X-rays, MRIs, and CT scans.
- Drug Discovery: AI speeds up the process of discovering new medicines.
- Personalized Treatment: AI suggests personalized treatment plans based on patient data.
- Examples:
  - IBM Watson Health
  - AI Wearables
  - UHG(United Health Group for Mediclaim)

EDUCATION

- **Personalized Learning:** AI adapts study material based on a student's performance.
- **Automated Grading:** AI can grade multiple-choice and written assignments.
- **Virtual Tutors:** AI-powered tutors assist students outside the classroom.
- **Examples:**
  - **Duolingo**.
  - **Coursera & edX**

AUTOMOBILE INDUSTRY

- **Autonomous Driving:** AI helps cars drive themselves with minimal human input.
- **Route Planning:** AI optimizes routes to reduce traffic and fuel consumption.
- **Driver Assistance:** AI-based systems prevent accidents by alerting drivers.
- **Examples:**
  - **Tesla Autopilot**
  - **Google's self-driving car project**

MANUFACTURING AND ROBOTICS

- **Predictive Maintenance:** AI predicts when machines will need repairs.
- **Robotic Automation:** AI controls robots to assemble products efficiently.
- **Quality Control:** AI inspects products for defects.
- **Examples:**
  - **BMW:** Uses AI-powered robots in car assembly.
  - **Coca-Cola:** Uses AI to manage bottling and packaging lines.

AGRICULTURE

- **Crop Monitoring:** AI analyzes soil health and crop growth using drones and sensors.
- **Pest Detection:** AI quickly identifies pest infections to protect crops.
- **Smart Irrigation:** AI helps in precise water management.

Examples:

- **AI Drones:** Monitor large farms to detect plant diseases.
- **Smart Irrigation Systems:** Adjust water supply based on crop needs.

ENTERTAINMENT AND MEDIA

- **Content Recommendation:** AI suggests songs, videos, and movies based on user preferences.
- **Content Creation:** AI writes articles, composes music, and generates video content.
- **User Personalization:** AI customizes entertainment experiences.
- **Examples:**
  - **Netflix & Spotify:** Recommend movies and songs you are likely to enjoy.
  - **YouTube:** Suggests videos based on your watch history.

RETAIL AND E-COMMERCE

- **Product Recommendations:** AI suggests products based on browsing history.
- **Customer Support:** AI-powered chatbots provide instant help to customers.
- **Inventory Management:** AI predicts product demand to maintain stock levels.

Examples:

- **Amazon & Flipkart:** AI recommends products based on customer behavior.
- **Myntra Chatbot:** Assists customers 24/7 with queries and order tracking.

RELEVANCE OF AI TO MAJOR ENGINEERING DISCIPLINES

- **Electrical / Electronics Engineering**
  - **Predictive Maintenance-** AI algorithms analyze data from sensors on electrical equipment (like motors, generators, and transformers) to predict when maintenance is needed.
  - **Intelligent Control Systems:**AI-powered control systems are being used in various applications, including robotics, industrial automation, and renewable energy systems.
    - For example, AI can optimize the performance of wind turbines by adjusting blade angles for maximum energy capture.
  - **Power Quality Improvement**
  - **AI in Electric Vehicles**

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RELEVANCE OF AI TO MAJOR ENGINEERING DISCIPLINES

- **Electrical Engineering -** Smart grid, power system automation –
- **Example**: Power Quality Improvement
- **Steps**:
- 1. Detect voltage/frequency faults via sensors
- 2. Input to neural network/expert system
- 3. Diagnose fault type
- 4. Trigger corrective action
- 5. Improve stability and reduce outages

RELEVANCE OF AI TO MAJOR ENGINEERING DISCIPLINES

- **Civil Engineering –**
- Traffic management, construction planning –
- **Example:**
- **Smart Traffic Signals –**
- **Steps:**
- 1. Live traffic data from cameras
- 2. AI processes vehicle flow
- 3. Adjusts signal timing dynamically
- 4. Reduces congestion
- 5. Learns traffic trends

RELEVANCE OF AI TO MAJOR ENGINEERING DISCIPLINES

Aerospace Engineering

Relevance: Autonomous navigation, flight safety

Example: Drone Navigation

Steps:

1. Sensors detect environment
2. AI processes GPS and visual data
3. Plans safest flight route
4. Avoids obstacles

RELEVANCE OF AI TO MAJOR ENGINEERING DISCIPLINES

- **Mechanical Engineering**
  - **Design Optimization**
  - **Automation and Robotics**
  - Maintenance prediction, automation –

Example: Predictive Maintenance in Factories –

Steps:

1. Sensors collect vibration/temperature data
2. AI model analyzes for failure patterns
3. Alert maintenance team
4. Fix before failure occurs
5. Increase machine life

RELEVANCE OF AI TO MAJOR ENGINEERING DISCIPLINES

- **Industrial Engineering** -Supply chain, inventory, production –
- **Example**: Demand Forecasting
- **Steps**:
- 1. Collect sales data
- 2. Use AI to predict demand
- 3. Automate reordering
- 4. Avoid over/under stocking

AGENT AND ENVIRONMENT

- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and acting upon that environment through **actuators.**
- *An agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived*

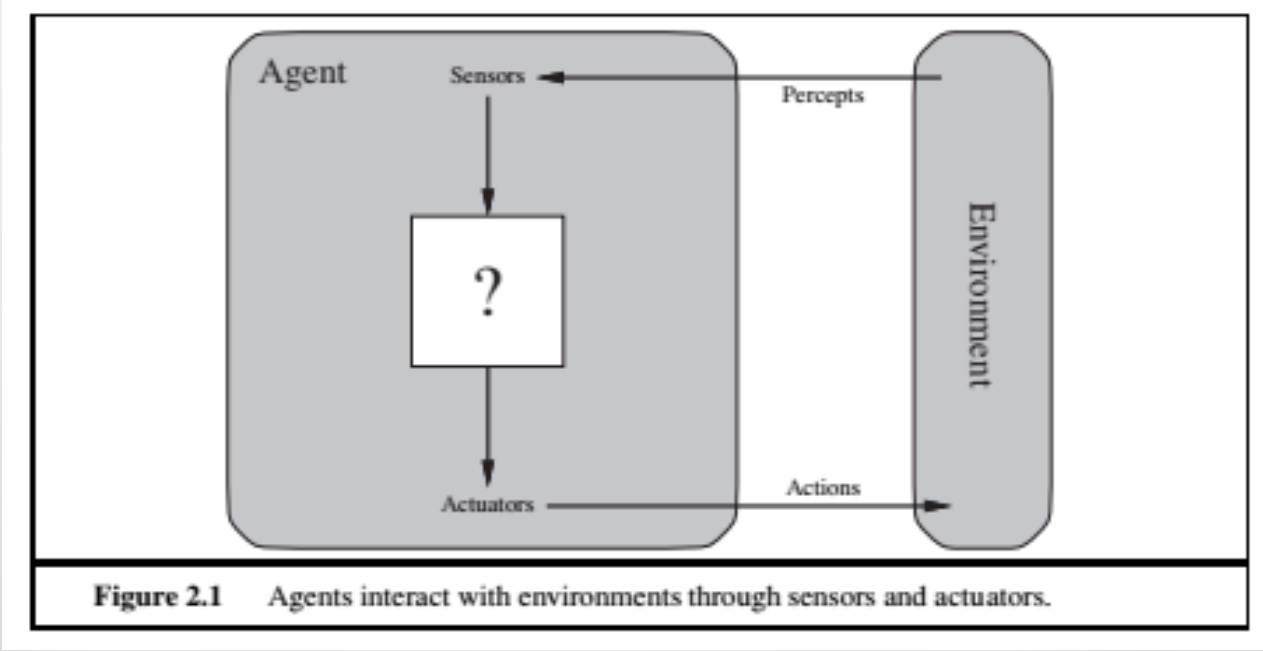


Figure 2.1 Agents interact with environments through sensors and actuators.

AGENT AND ENVIRONMENT

- The **agent function** is an abstract mathematical description; the **agent program** is a concrete implementation, running within some physical system.

A **rational agent** is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly.

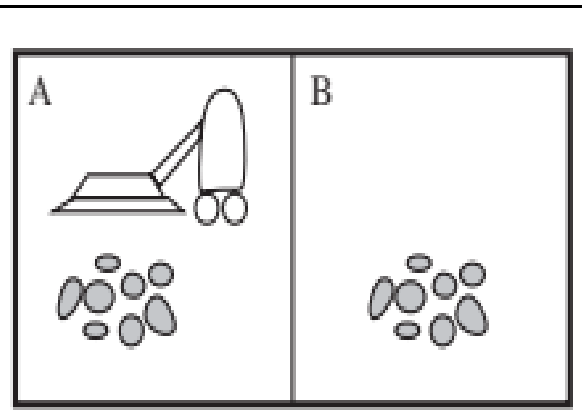


Figure 2.2 A vacuum-cleaner world with just two locations.

RATIONAL AGENT

- What is rational at any given time depends on four things:
- • The performance measure that defines the criterion of success.
- • The agent's prior knowledge of the environment.
- • The actions that the agent can perform.
- • The agent's percept sequence to date.

- *For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.*

SPECIFYING THE TASK ENVIRONMENT

P- Performance
E- Environment
A- Actuators
S - Sensors)

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

PEAS

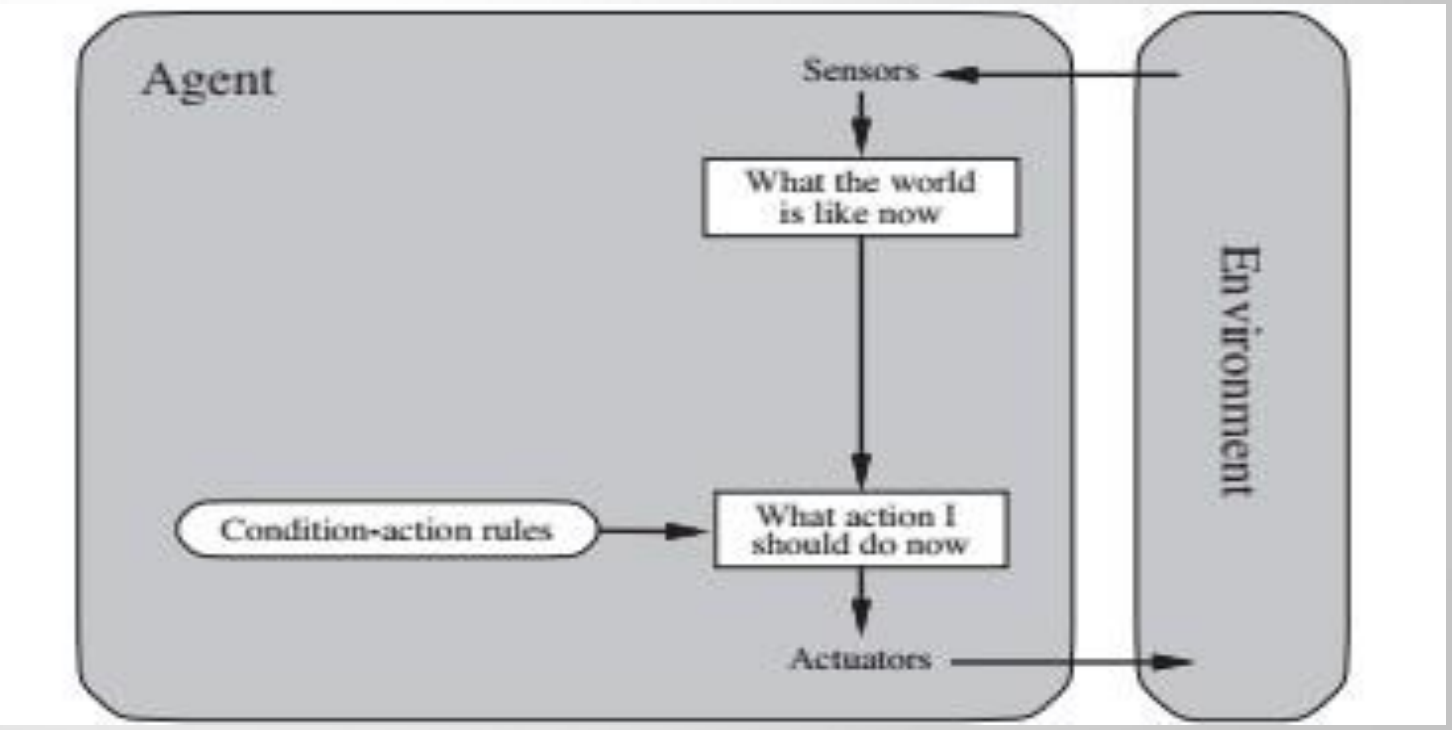
Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

TYPES OF AGENT

1. Simple reflex agents
2. Model-based reflex agent
3. Goal-based agents
4. Utility Based agents

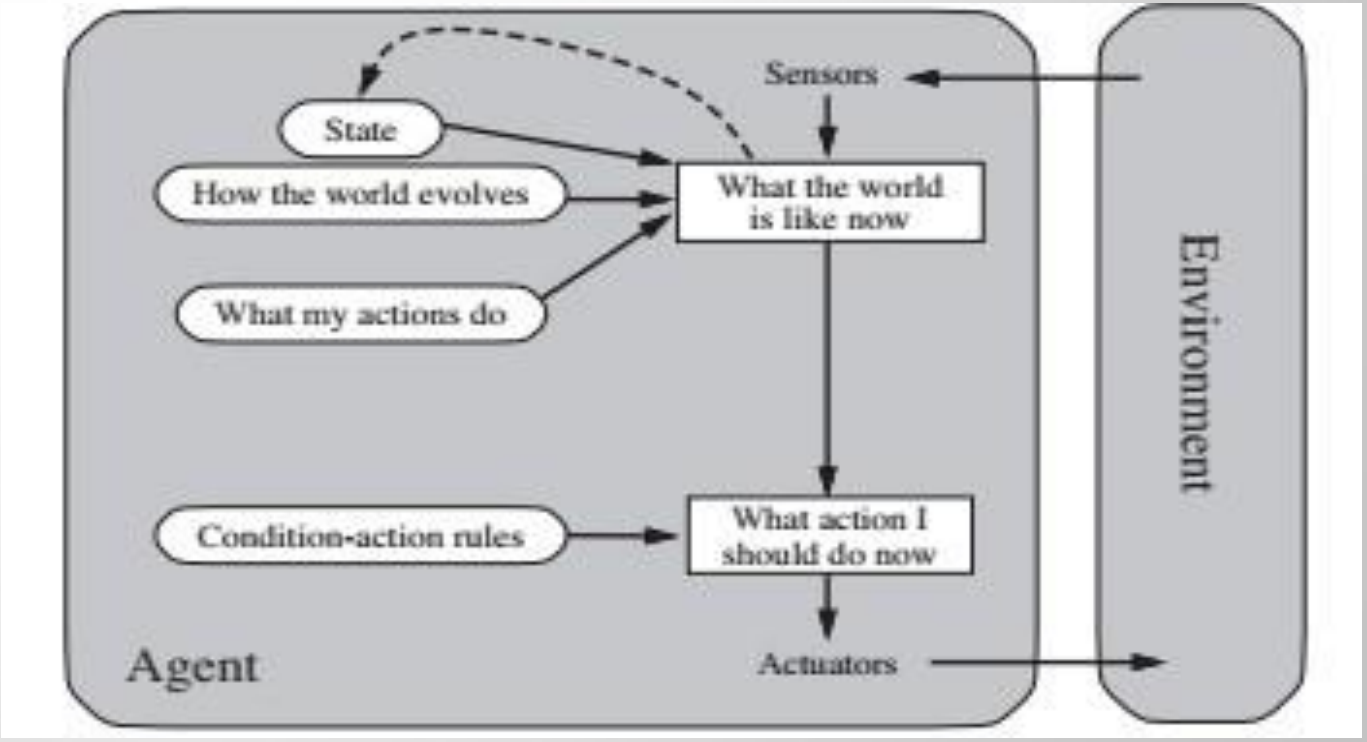
SIMPLE REFLEX AGENTS

- **Simple reflex agents**
  These agents select actions on the basis of the *current* percept, ignoring the rest of the percept history.

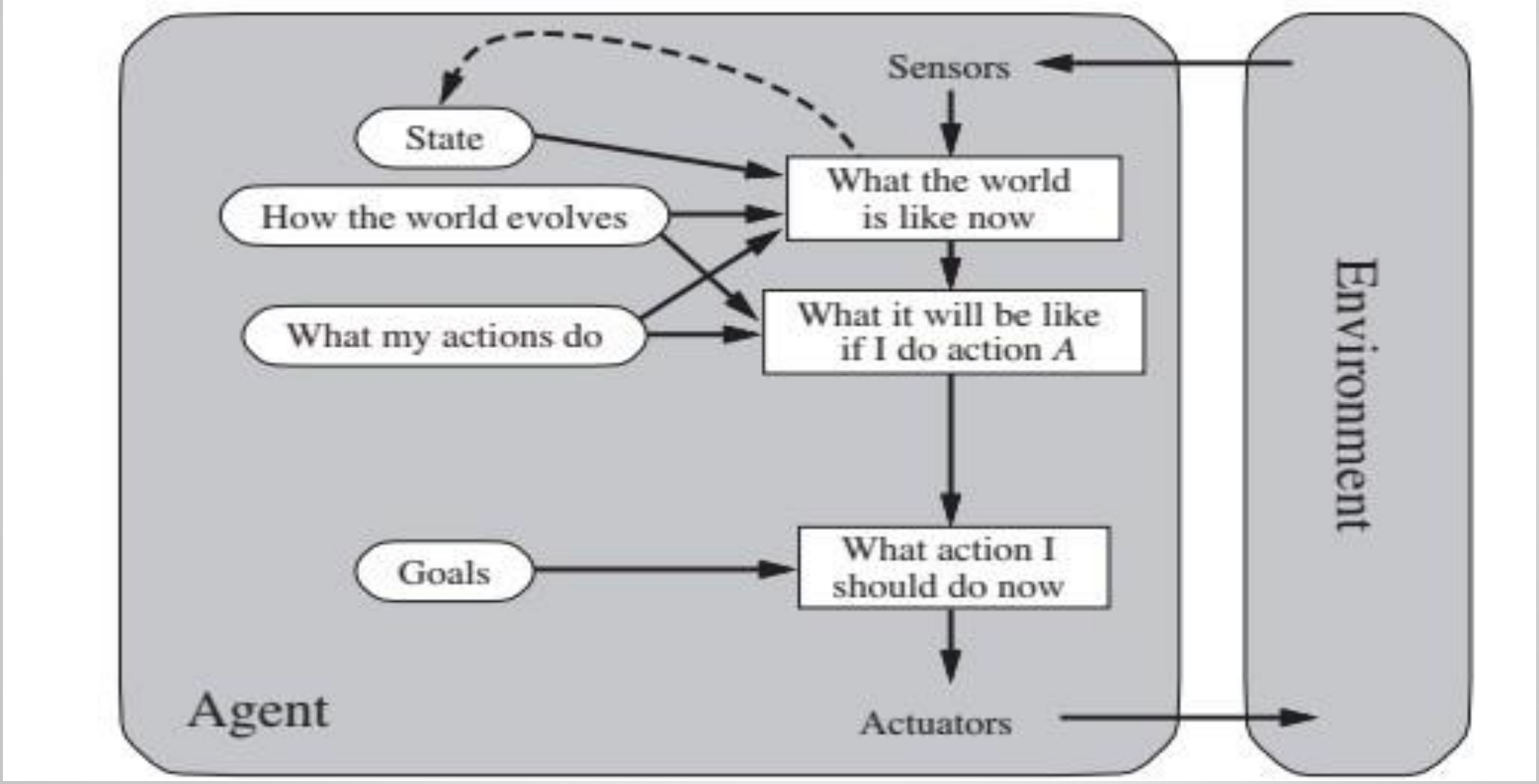


MODEL-BASED REFLEX AGENT

- The most effective way to handle partial observability is for the agent to *keep track of the part of the world it can't see now*. That is, the agent should maintain some sort of **internal state** that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.



GOAL-BASED AGENTS



UTILITY-BASED AGENTS

